

7.2.11

AI25BTECH11017-BALU

Question:

Find the equation of the circle having $(1, -2)$ as its centre and passing through the intersection of

$$3x + y = 14, \quad 2x + 5y = 18. \quad (0.1)$$

Solution:

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,

$$\begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 14 \\ 18 \end{pmatrix} \quad (0.2)$$

$$\mathbf{B} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \quad (0.3)$$

circle equation is

$$\mathbf{xx}^T + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (0.4)$$

$$\mathbf{u} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (0.5)$$

point $\mathbf{B}$ lies on circle,so

$$\mathbf{bb}^T + 2\mathbf{u}^T \mathbf{b} + f = 0 \quad (0.6)$$

$$f = -20 \quad (0.7)$$

hence the requiried circle equation is

$$\mathbf{xx}^T + 2 \begin{pmatrix} 1 & -2 \end{pmatrix} \mathbf{x} - 20 = 0 \quad (0.8)$$

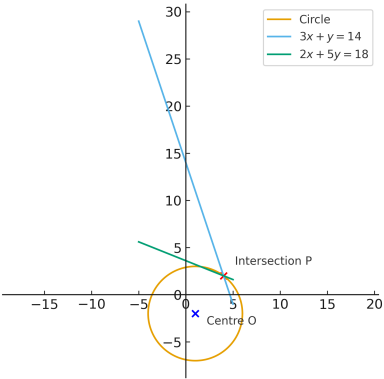


Fig. 0.1